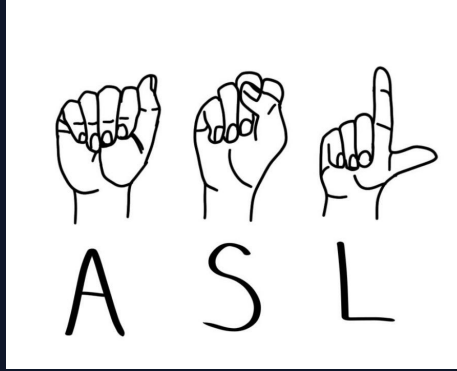


GenASL



AI-Powered ASL Video Overlay for YouTube

Proof of Concept → Business Context, Findings & Recommendations

The Accessibility Gap

 466M

people worldwide with
disabling hearing loss

 2-5%

of online video has
sign language interpretation

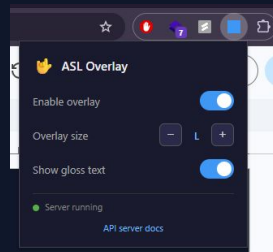
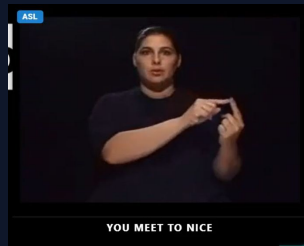
 

Auto-captions $\neq$ ASL
ASL is a distinct language

Opportunity

An AI-powered ASL overlay that works on existing YouTube videos - zero content creator effort required. GenASL bridges the gap between auto-captions and true sign language accessibility.

How GenASL Works



1

Transcript Ingestion

Fetch & normalize
YouTube captions

2

LLM Gloss Translation

English → ASL
gloss via LLM

GenAI

Ollama (local) · Gemini · OpenAI

3

Clip Lookup & Chaining

Match glosses → WLASL
clips → FFmpeg concat

4

Real-Time Overlay

Chrome extension
PiP on YouTube

Example

YouTube Caption

"What is your name?"

LLM Gloss Output

YOUR NAME WHAT

WLASL Clips

3 clips chained
(YOUR + NAME + WHAT)

Overlay Result

ASL video playing
in PiP on YouTube

Key Technical Findings



End-to-End Pipeline Working

Transcript → LLM Gloss → Video Clips → Real-time overlay. Full chain operational.



Provider-Agnostic LLM Design

Supports Ollama (local), Google Gemini, and OpenAI models APIs



~10× Speedup via Batching

Single transcript fetch + chunked translation (10 lines/call) vs. per-caption API calls.



Confidence Gate at 0.80

Cosine similarity threshold acts as allowlist - unmatched glosses fall back to captions.

Governance & Responsible AI



Transparency

"AI-generated ASL overlay (POC)" label burned into output - non-removable.



Accuracy Safeguard

0.80 confidence threshold for English -> ASL Gloss word matching



Cultural Sensitivity

Word-level glosses disclosed - not grammatically correct ASL. No DHH community review yet (accepted limitation).



Privacy & Compliance

Anonymous feedback, no PII retained, local-only cache. WLASL under academic license.

Risk Matrix

Risk	Status	Mitigation / Notes
Incorrect ASL match shown	Mitigated	0.80 threshold + bracket stripping
Word-level clips misrepresent ASL grammar	Accepted	Documented in all user-facing labels
No Deaf community review	Accepted	Recommended for next phase
LLM hallucination in gloss	Mitigated	Regex sanitization, 10-line batching, fallback
Overlapping clip timestamps	Mitigated	Overlap resolution: higher-scoring entry kept



Let us show you the system in action →

<https://github.com/sanaro99/genasl>

Go / No-Go Recommendation

GO

with conditions

Rationale

The POC demonstrates that real-time AI-powered ASL overlay on YouTube is technically feasible. The pipeline is functional, the multi-provider LLM architecture provides flexibility, and governance controls are in place.

Conditions for Go



Conduct Deaf community user testing before any production deployment



Address ASL grammar reordering (currently word-level only)



Migrate from youtube-transcript-api to official YouTube Data API v3

Success Factors & Governance



DHH Community Co-Design

Validate gloss quality and overlay UX with Deaf users and professional interpreters



ASL Grammar Layer

Add topic-comment reordering, spatial referencing, non-manual markers



Vocabulary Expansion

Expand beyond 2,000 glosses - add medical, legal, education domains



LLM Quality Assurance

Automated benchmarks (BLEU / human eval) against certified interpreter transcripts

Required Governance Controls



Threshold changes require review, documentation, and approval



"AI-generated ASL" disclosure label must be non-removable



Medical/legal/emergency content out of scope until validated



Regular bias & fairness audits across dialects



Migrate to YouTube Data API v3 before non-academic use

Roadmap & Next Steps

IMMEDIATE

Recruit DHH participants for usability testing. Collect overlay quality and disclosure feedback.

SHORT-TERM

LLM-based ASL grammar reordering (topic-comment). Expand WLASL vocabulary to 5,000+ glosses.

MEDIUM-TERM

Migrate to YouTube Data API v3. Build automated gloss quality regression. Explore 3D avatar rendering.

LONG-TERM

Partner with Deaf orgs for co-design. Evaluate production deployment (cloud vs. browser-local).